



Multilingual Entity and Relation Extraction from Unified to Language-specific Training

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ABSTRACT

Entity and relation extraction is a key task in information extraction, where the output can be used for downstream NLP tasks. Existing approaches for entity and relation extraction task mainly focus on the English corpora and ignore other languages. Thus, it is critical to improve performance in a multilingual setting. Meanwhile, multilingual training is usually used to boost cross-lingual performance by transferring knowledge from languages (e.g., high-resource) to other (e.g., low-resource) languages. However, language interference usually exists in multilingual tasks as the model parameters are shared among all languages. In this paper, we propose a two-stage multilingual training method and a joint model called Multilingual Entity and Relation Extraction framework (mERE) to mitigate language interference across languages. Specifically, we randomly concatenate sentences in different languages to train a Language-universal Aggregator (LA), which narrows the distance of embedding representations by obtaining the unified language representation. Then, we separate parameters to mitigate interference via tuning a Language-specific Switcher (LS), which includes several independent submodules to refine the language-specific feature representation. After that, to enhance the relational triple extraction, the sentence representations concatenated with the relation

feature are used to recognize the entities. Extensive experimental results show that our method outperforms both the monolingual and multilingual baseline methods. Besides, we also perform detailed analysis to show that mERE is lightweight but effective on relational triple extraction and mERE is easy to transfer to other backbone models of multi-field tasks, which further demonstrates the effectiveness of our method.

CCS CONCEPTS

• Computing methodologies → Information extraction.

KEYWORDS

Joint extraction, Information extraction, Multilingual entity and relation extraction, Relational triple extraction

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1 INTRODUCTION

Entity and relation extraction (ERE) contains two sub-tasks called named entity recognition (NER) [8, 10, 21, 23] and relation classification (RC) [13, 14], which is the fundamental step of automatic knowledge graphs (KGs) [15] construction, knowledge discovery and intelligent question answering system. The results of ERE are typically described as a relational triple (h, r, t) , where h and t are the head entity and the tail entity, respectively, and r denotes the relation between them. For example, for the sentence “Big Ben

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is in UK.” with a predefined relation called “Locate_in”, an ideal relational triple of this sentence is expressed as (Big Ben, Locate_in, UK).

As a large amount of data is available from different languages on the Internet, it is important to utilize such valuable resources and develop multilingual entity and relation extraction models, which can operate across language barriers. However, most existing methods propose to solve ERE task using English corpora, which can only deal with the monolingual extraction task. The main reason is that many languages suffer from the scarcity of corpora in ERE task. Thus, multilingual training is proposed to help each other in a shared model, where the well-trained knowledge of high-resource languages can be transferred to low-resource languages with a small amount of data. Recently, [19] proposed a multilingual dataset called SMiLER, which is the first work to apply both monolingual and multilingual training. The authors in [19] introduce the multilingual entity and relation extraction model (i.e., HERBERTa) without considering interference across languages. However, such language interference is prevalent in multilingual tasks because of parameter sharing [6, 24, 28]. As shown in Figure 1, the four sentences express the same meaning in different languages. To mitigate interference among languages, we propose to extract the feature representation of the corresponding language sentence. First, to facilitate the cross-lingual transfer among different languages, multilingual representations are supposed to be closed under similar semantics using cross-lingual sentence-level concatenation. Then, based on the shared multilingual parameters, the language-specific representations derived from the independent modules can mitigate interference among multiple languages. Specifically, we propose a two-stage multilingual training method

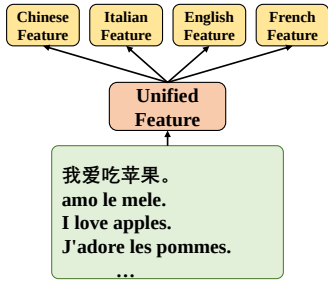


Figure 1: The overview of multilingual sentences from unified feature representation to language-specific feature representations.

and an effective model called multilingual Entity and Relation Extraction framework (mERE) to address the multilingual ERE task. In the first stage, we utilize a cross-lingual encoder to encode different language sentences and extract relations directly. Then, we train the joint model with our Language-universal Aggregator (LA) to generate the unified language feature, which narrows the distance of similar semantic representation across languages. LA consists of a self-attention layer and is trained by random multi-sentence concatenation, which is used to learn semantic similarities in multilingual training. In the second stage, to alleviate the interference among languages, we freeze the parameters of LA and cross-lingual

encoder in the first stage and optimize the independent parameters via fine-tuning the model with a Language-specific Switcher (LS), which consists of several independent submodules to produce the specific language features. Meanwhile, a selection mechanism is applied to choose the optimal group of submodules from LS, which enables the submodule to share the same parameters with a certain group of languages. Such an automatic selection mechanism saves many model parameters when the number of languages is large. After that, each token representation is concatenated with the relation representation to enhance the recognition of the positions of entities in a sentence. Finally, in mERE, we adopt joint training to mitigate the error propagation problem.

We conducted extensive experiments on the SMiLER benchmark of 14 languages with 36 relations (including no_relation) in total. The experimental results demonstrate that our method outperforms previous monolingual and multilingual ERE baseline methods by a large margin across languages, which demonstrates that our method can effectively mitigate language interference by improving representation quality among languages. Besides, we conduct detailed experiments to analyze how our method affects relational triple extraction. Moreover, our method is simple but effective, and it is also easy to transfer to different backbone models of multi-field tasks with lightweight modules.

2 RELATED WORK

2.1 Entity and Relation Extraction

Early entity and relation extraction task use a pipeline approach, which are two separate sub-tasks including named entity recognition and relation classification. [20] first works on Recurrent Neural Network (RNN) based model for extraction, capturing the semantics of the entity and its adjacent phrases through parsing trees. Most studies focus on the joint approach, which models entity recognition and relation classification in the same network and naturally relieves error propagation problem. [12] proposes a multi-feature fusion sentence representation and decoder sequence annotation to handle the overlapping triples that are overlapped with one or two entities. Another architecture is joint decoding, which extracts entity pairs and corresponding relations simultaneously in one stage.

2.2 Multilingual Models

Multilingual models are a type of model that performs cross-lingual transfer among different languages, such as multilingual pre-training [2, 3, 17, 25, 27] and machine translation [7, 26, 28, 29]. Specifically, mBERT pre-trained on 104 languages in Wikipedia has a strong ability for cross-lingual transfer. Multilingual neural machine translation (MNMT) trains a single NMT model in multiple language pairs supporting translation directions between multiple languages by sharing parameters [4, 5, 9, 16]. Early studies mainly utilize high-resource languages to help low-resource languages and even perform zero-shot transfer translation [1, 30]. Recent studies focus on designing language-specific components to mitigate the language interference in shared parameters, especially on high-resource pairs [18, 22, 28]. Our method boosts the sentence representation quality from superior unified representation to further

language-specific representation. Existing entity and relation extraction datasets are insufficient in diversity and size. English is always used to be training corpora. [19] presents a new, large, and diversified dataset Samsung Multilingual Entity and Relation Extraction (SMILER) dataset to entity and relation extraction both for English and multilingual settings.

3 METHODOLOGY

In this section, we introduce the details of our training method for the multilingual joint entity and relation extraction framework. We propose a two-stage training strategy, as shown in Figure 2 left part. In the first stage, we train a Language-universal Aggregator (LA) for learning the unified representations among multiple languages. In the second stage, we freeze the parameters and fine-tune the Language-specific Switcher (LS), which is applied to select specific feature representations of various languages.

3.1 Task Formulation

The goal of the multilingual joint entity and relation extraction aims to identify all possible relational triples from sentences in different languages. Formally, given a sentence X from multilingual corpora $D = \{D_n\}_{n=1}^N$, where N represents the number of the all languages $L_{all} = \{L_n\}_{n=1}^N$. The probability of the target triple $Y = \{s, r, o\}$ is defined as below:

$$P(Y | X) = p(r | X; \phi) p(s, o | X, r; \phi), \quad (1)$$

where r denotes relation, s and o are subject (head entity) and object (tail entity), respectively. $p(r | X; \phi)$ means relation is only related to sentence X , and $p(s, o | X, r; \phi)$ means the entity pair (s, o) is related to both sentence X and the relation r that they shared.

3.2 Language-aggregation Training

We train the model with a Language-universal Aggregator (LA) to learn the unified representation, which effectively narrows the distance of semantic representations across different languages. To obtain context representations of each token from the multilingual sentences, we utilize the cross-lingual pre-trained encoder for building a multilingual model. Given the sentence $X^{L_n} = \{x_1^{L_n}, \dots, x_i^{L_n}, \dots, x_m^{L_n}\}$ with m tokens (including [CLS], [SEP] and [PAD]), $x_i^{L_n} \in \mathbb{R}^d$ is the i -th token embedding and d is the embedding size. The whole sentence is encoded by the cross-lingual pre-trained encoder:

$$h^{L_n} = \mathcal{H}(X^{L_n}; \phi), \quad (2)$$

where $h^{L_n} = \{h_1^{L_n}, \dots, h_i^{L_n}, \dots, h_m^{L_n}\} \in \mathbb{R}^{m \times d}$ represents the encoded representation and d is the hidden size. $\mathcal{H}$ denotes the cross-lingual pre-trained encoder. Meanwhile, a relation classifier $W^r \in \mathbb{R}^{d \times U}$ is used to project pooled output vector h_p (from the [CLS] token) to the relation r_c , where U is the number of relation types. The relation extraction is defined as:

$$r_c = h_p W^r, \quad (3)$$

To better learn the unified semantic representation among multiple languages, we randomly sample s sentences of different languages from the training corpora to generate the cross-lingual representations using Equation (2) and concatenate them to obtain

$h_{cat} = [h_1^{L_{X_1}}, \dots, h_i^{L_{X_i}}, \dots, h_s^{L_{X_s}}]$, where L_{X_i} denotes the language symbol of the i -th sentence. Considering that each token needs to capture the dependency of inner-sentence and acquire semantic similarity representation of inter-sentence among languages, we train LA which applies the self-attention mechanism for fusing the information of the given concatenated representation:

$$\hat{h}_{cat} = \text{SF}\left(\frac{QK^T}{\sqrt{\epsilon}}\right)V \quad (4)$$

where $Q = h_{cat} W_q$, $K = h_{cat} W_k$ and $V = h_{cat} W_v$. SF represents the softmax operation. The three-parameter matrices W_q , W_k , and W_v are trainable. The term $1/\sqrt{\epsilon}$ is the scaling factor. $\hat{h}_{cat} = \{\hat{h}_1^{L_{X_1}}, \dots, \hat{h}_i^{L_{X_i}}, \dots, \hat{h}_s^{L_{X_s}}\}$ and $\hat{h}_i^{L_{X_i}}$ is i -th element. Instead of using language-specific features generated via Equation (8), we directly utilize each element representation in $\hat{h}_i^{L_{X_i}}$ to train the model via Equation (9). Figure 2 right part shows that LA is trained with a concatenation of two random sentence representations, which are illustrated as the green boxes (English) and yellow boxes (Italian).

3.3 Language-specific Training

To acquire features of a specific language, we freeze the parameters of LA and cross-lingual encoder in the first training stage and fine-tune the model with LS. After obtaining the unified representation via LA, we extract the language-specific features via the LS with the selection mechanism from the unified representations.

Given the language symbol $L_n \in L_{all}$ ($1 \leq n \leq N$) and our LS $\theta = \{\theta_t\}_{t=1}^{\mathcal{T}}$ ($1 \leq t \leq \mathcal{T}$, $1 \leq \mathcal{T} \leq N$), our selection mechanism is used to select corresponding submodules $\theta_{f(L_n)}$, in which $f(\cdot)$ is a function that maps a language to corresponding LS modules. To design an appropriate map function for our selection mechanism, each sentence is prefixed to the corresponding language symbol, which enables the model to correctly route sentences. Besides, all submodules from LS attend to the selection procedure during the training stage, which solves the undifferentiability problem. Specifically, the function $f_t(\cdot)$ indicates the probability of selection of submodule θ_t :

$$f_t(L_n) = \frac{\exp(e_t^{L_n})}{\sum_{i=1}^{\mathcal{T}} \exp(e_i^{L_n})} \quad (5)$$

where $e_i^{L_n}$ is i -th element of the probability vector $e^{L_n} = E_l[n] W_f$. $E_l \in \mathbb{R}^{N \times d}$ denotes the look-up table for all language prefix embeddings. The router matrix $W_f \in \mathbb{R}^{d \times \mathcal{T}}$ is used to project e^{L_n} which are normalized via a softmax distribution over the total $\mathcal{T}$ modules.

For each submodule θ_t from θ , we utilize $\mathcal{E}_{\theta_t}(\cdot)$ to transform unified feature representation $\hat{h}^{L_n}$ into language-specific feature branch $\hat{h}_{\theta_t}^{L_n}$:

$$\hat{h}_{\theta_t}^{L_n} = \mathcal{E}_{\theta_t}(\hat{h}^{L_n}) \quad (6)$$

$$\mathcal{E}_{\theta_t}(\hat{h}^{L_n}) = \text{LN}\left(\sigma(\hat{h}^{L_n} W_u) W_d + \hat{h}^{L_n}\right) \quad (7)$$

where $\hat{h}^{L_n} \in \mathbb{R}^{m \times d}$ is an element of $\hat{h}_{cat}$. $W_u \in \mathbb{R}^{d \times b}$ and $W_d \in \mathbb{R}^{b \times d}$ are projection matrices ($b > d$). σ is the ReLU activation function and $\text{LN}(\cdot)$ is the layer normalization function.

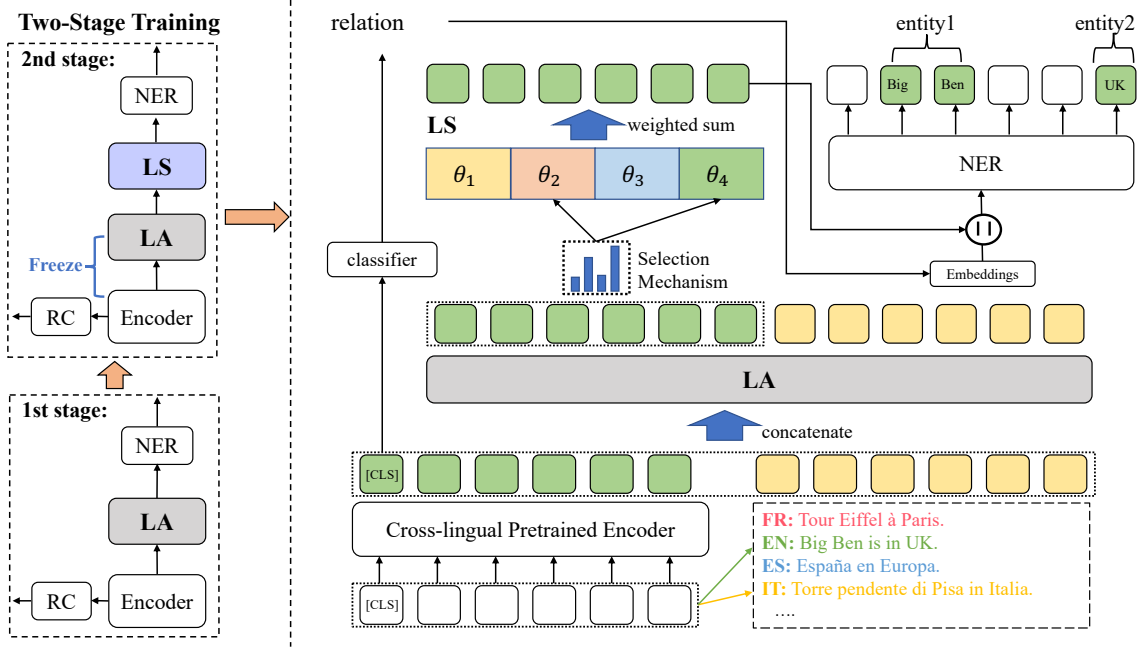


Figure 2: The overview of mERE.

To ensure gradients are propagated to all submodules of LS $\{\theta_t\}_{t=1}^T$, we apply the weighted average for obtaining the language-specific feature:

$$\tilde{h}^{L_n} = \sum_{t=1}^T f_t(L_n) \mathcal{E}_{\theta_t}(\hat{h}^{L_n}) \quad (8)$$

For the whole process, function $f_t(L_n)$ in Equation (8) permits differentiability of the router.

In the evaluation stage, it is necessary to prune several branches with the lowest selection probabilities to obtain the best performance. Therefore, we use the top- k strategy to select the best k ($1 \leq k \leq T$) submodules with the highest probabilities to generate the language-specific representation. When $k = T$ indicates all submodules involved in the calculation which means the selection mechanism is the same as the training stage. The mapping process is described as: $L_n \rightarrow \{\pi_1^{L_n}, \dots, \pi_i^{L_n}, \dots, \pi_k^{L_n}\} \in \Pi(k)$, where $\pi_i^{L_n}$ is one of the submodule index that corresponds to language L_n and $\Pi(k)$ is the space of all k -length combinations of $C_{\mathcal{T}}^k$ in total. Figure 2 right part shows two submodules with the highest probabilities are selected from LS (four submodules in total as illustrated) during the evaluation stage.

3.4 Entity Recognition

After obtaining the language-specific representation from LS, we create four matrices to recognize the head and tail positions of two named entities. To enhance the accuracy of recognition, we add a relation feature that constrains the extracted entities that are only related to the relevant relation. Formally, given a language-specific representation $\tilde{h}^{L_n} \in \mathbb{R}^{m \times d}$ of the m -length sentence and the relation vector r_e retrieved from relation embedding table $E_r \in \mathbb{R}^{I \times d}$,

where I is the number of relations, the two entities are recognized as followed:

$$entity^x = (\eta((\tilde{h}^{L_n} \oplus r_e)W_y))U_y \quad (9)$$

where the symbol collection $entity = \{\text{head}, \text{tail}\}$, $x = \{\text{start}, \text{end}\}$ and $y = \{hs, he, ts, te\}$. We concatenate the relation vector with each token representation to enhance the recognition of entities, namely $\tilde{h}^{L_n} \oplus r_e = \{[\tilde{h}_1^{L_n}, r_e], \dots, [\tilde{h}_i^{L_n}, r_e], \dots, [\tilde{h}_m^{L_n}, r_e]\} \in \mathbb{R}^{m \times 2d}$. $W_y \in \mathbb{R}^{2d \times d}$ are four down projection matrices and $U_y \in \mathbb{R}^{d \times 1}$ are four index projection matrices. η denotes $\tanh$ activation function.

3.5 Training Objective

Our model presented in Figure 2 is trained jointly on multilingual ERE corpora. We first train the model only using a multilingual training strategy for our Language-universal Aggregator. Based on the unified language representation, we fine-tune the model with Language-specific Switcher for learning the language-specific feature in the next step. The objective is to minimize the two training loss functions which are defined below:

$$\mathcal{L}_{LAT} = \sum_{m=1}^M \mathbb{E}_{(x,y) \sim D_m} [\mathcal{L}_{ere}(x, y; \Theta)] \quad (10)$$

$$\mathcal{L}_{LST} = \sum_{m=1}^M \mathbb{E}_{(x,y) \sim D_m} [\mathcal{L}_{ere}(x, y; \Theta, \theta)] \quad (11)$$

where D means multilingual entity and relation extraction training corpora and M denotes the number of the samples. Θ indicates shared parameters and θ is parameters in LS with selection mechanism. $\mathcal{L}_{ere}$ is the loss function for entity and relation extraction,

which is defined as below:

$$\mathcal{L}_{ere} = \frac{\alpha}{2}(\mathcal{L}_h^{\text{start}} + \mathcal{L}_h^{\text{end}} + \mathcal{L}_t^{\text{start}} + \mathcal{L}_t^{\text{end}}) + \beta\mathcal{L}^{\text{rel}} \quad (12)$$

where each $\mathcal{L}$ with any superscript is a cross-entropy loss. The subscripts with h and t indicate the head entity and tail entity respectively. The *start* and *end* of superscripts denote the first token index and last token index of an entity separately. $\mathcal{L}^{\text{rel}}$ is the loss function for relation classification. α and β are two weights on entity recognition loss and relation classification loss respectively.

4 EXPERIMENTS

4.1 Datasets

We evaluate mERE on the SMiLER test sets [19]. SMiLER is the largest and most diversified multilingual dataset for multilingual entity and relation extraction task. It consists of about 1.1M annotated sentences from Wikipedia and DBpedia, 36 relation types in total. The corpora include 14 languages: English (En), Korean (Ko), Italian (It), French (Fr), German (De), Portuguese (Pt), Netherlands (Nl), Polish (Pl), Spanish (Es), Arabic (Ar), Russian (Ru), Swedish (Sv), Farsi (Fa), Ukrainian (Uk). The statistics⁷ of SMiLER are shown in Table 1, languages are ordered from high-resource languages (left) to low-resource languages (right). English corpora include full-size, middle-size, and small-size. EN-small is treated as our English corpus in this study. As the development set in SMiLER is not publicly available, we only randomly extract the sentences from the training set to create new files with the same split ratio as the original paper.

4.2 Implementation Details

We utilize mBERT as our cross-lingual encoder. We train our model with AdamW, the learning rate is $3e-5$ and weight decay is 0.1. The batch size is set to 16 on the Tesla V100 GPU. The hidden size d is 768 and dimension b of projection matrices W_u and W_d is 1024. The max sequence length is 256 and we concatenate 2 sentences during the first training stage. The submodule number of LS $\mathcal{T}$ is set to 6 (2 layers for 3 submodules and 1 layer for the other 3 submodules) during the second training stage. The epoch of the first stage is set to 5. The max epoch of the second stage is set to 8 with an early stopping mechanism. The loss weight is set to 2 in named entity recognition and 1 in relation classification. In the evaluation stage, $k = 3$ in the top- k strategy.

4.3 Evaluation

We adopt standard micro- F_1 metric to calculate scores on the models. The extracted entity pair is regarded as correct if the predictions of the head entity and tail entity are both same with the ground truth. A triple is treated as correct if the entity pair and the corresponding relation type are all correct. *no_relation* type is included in relation prediction. We also add a mask for the relation types which are not existed in a certain language.

4.4 Baselines

As far as we know, the SMiLER is a new dataset and there is only one existing method for multilingual entity and relation extraction

task. Therefore, we reproduce the following competitive baselines to compare with our proposed approach for a fair comparison:

- HEBERTa [19]: A multilingual entity and relation extraction framework called Hybrid Entity and Relation extraction BERT, which achieves the state-of-the-art performance on SMiLER. HEBERTa uses a pipeline training manner that combines two independent BERT models.
- mBERT [11]: A cross-lingual model first uses the mBERT as a backbone for relation classification (RC), which is trained on 104 languages with the corresponding Wikipedia dumps. We reproduce the results with the code shared at <https://github.com/boun-tabi/RELX>
- MTMB [11]: A multilingual pre-training scheme called Matching the Multilingual Blanks (MTMB). The framework shows several advantages against the mBERT on monolingual tasks and achieves significant improvements in cross-lingual transfer. Note that this framework is only designed for RC. Therefore, we modified the output layer of the baseline to conduct the ERE task.

4.5 Experimental Results and Analysis

We set four language groups to analyze the impact of languages: (1)MONO: 14 languages in monolingual training. (2)EURO: European languages which include It, Fr, Pt, De, Es, En. (3)SVO¹: EURO, Ru, Sv, Nl, Pl, Uk. (4)The default group is all language corpora involved and training together.

The main experiment results are shown in Table 2. The symbol - in the data results denotes that the language is not involved under the corresponding language group. AVG means the average F_1 score across all languages. From Table 2, our method improves multilingual model by a large margin over previous baselines. There is a 2.3% F_1 value improvement on average F_1 score compared with the previous strongest baseline MTMB. mERE achieves the best performance in 8 out of 14 languages, especially in high-resource languages. It also achieves the second-best performance in the other 5 out of 6 languages. Surprisingly, even our baseline mERE-LS-LA has 0.9% improvement over the MTMB. It seems that our basic structure is more effective on multilingual entity and relation extraction task. Compared with mERE-LS that only uses LA, our full model mERE has nearly 0.7% F_1 value improvement on average and yields similar or higher results in 13 languages except for Sv. The improvement can be attributed to our switcher-based language-specific training strategy, which finally extracts accurate information for entity recognition in each language. Compared with our baseline mERE-LS-LA, our full model mERE has nearly 1.4% F_1 value improvement on average, that means mERE-LS also has nearly 0.7% F_1 value improvement on average. All such impressive results demonstrate that our full model mERE truly enhances the representation quality and mitigates language interference to a certain extent.

¹SVO stands for the relative position of the Subject, Verb, and Object in the typical affirmative sentence. We treat Korean, Farsi, and Arabic as non-SVO languages. Arabic is VSO, while Korean and Farsi are SOV.

²Models are reproduced by us in our experiment settings. Note that we modified the output layer of mBERT and MTMB to implement the entity recognition to accommodate the ERE task and train the model in the joint training method.

Table 1: The statistics of SMiLER dataset.

Languages	EN-full	EN-mid	It	Fr	De	Pt	Nl	En-small	Ko	Pl	Es	Ar	Ru	Sv	Fa	Uk
sentences num.	748k	269k	76k	62k	53k	45k	40k	35k	20k	17k	12k	9k	7k	5k	3k	1k
relation types	36	36	22	22	22	22	22	32	28	22	22	9	8	22	8	7

Table 2: The F_1 scores of different models in 14 languages.

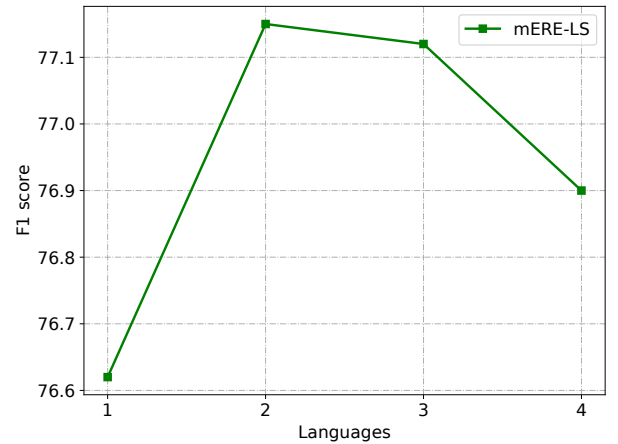
Models	AVG	It	Fr	De	Pt	Nl	En	Ko	Pl	Es	Ar	Ru	Sv	Fa	Uk
HERBERTa ²	75.5	83.9	68.7	71.5	72.1	78.5	60.9	80.4	83.1	60.0	88.4	79.4	84.8	79.6	65.0
mBERT ²	75.2	81.5	68.2	70.7	71.0	77.6	59.9	78.5	81.1	61.3	89.5	81.7	81.5	79.6	70.0
MTMB ²	75.6	80.9	67.8	70.9	70.3	79.1	58.3	79.3	82.2	58.2	91.1	74.1	83.7	77.8	85.0
mERE	77.9	81.7	70.3	73.4	74.3	81.1	62.3	82.7	81.6	64.7	91.6	83.1	83.7	79.6	80.0
mERE (EURO)	70.9	81.4	70.2	72.1	74.2	-	62.2	-	-	65.2	-	-	-	-	-
mERE (SVO)	75.7	81.3	70.0	72.9	73.3	80.6	62.1	-	81.0	64.7	-	83.1	83.7	-	80.0
mERE-LS	77.2	80.9	69.7	72.0	73.5	80.4	62.2	80.4	81.6	62.1	91.6	83.1	84.8	77.8	80.0
mERE-LS-LA (MONO)	70.9	81.2	68.3	67.1	68.4	77.9	58.6	79.3	79.0	48.4	90.0	72.5	80.4	66.7	55.0
mERE-LS-LA	76.5	81.3	69.0	71.9	71.4	80.3	60.3	76.4	84.2	60.7	90.0	83.9	83.7	77.8	80.0

Compared with mERE-LS-LA training on multilingual corpora in default, we can observe that multilingual training achieves much higher results than mERE-LS-LA (MONO) monolingual training from Table 2, especially in low-resource languages. Such as improvements of Uk (+25%), Fa (+11.1%), and Ru (+11.4%). It demonstrates that languages with less training data can benefit most from high-resource languages in multilingual training including ERE task. Compared with mERE training on the EURO group, the performance is close to the default setting. It is worth noting that Es achieves the best score in the EURO group. We conclude that Es benefits a lot from similarities of languages that are in the same language family, even with less training data. In the SVO group, we can also visualize that most languages in EURO decrease slightly with the interference of other non-EURO languages. The big difference in syntactic structures might be the main interference among languages. However, compared with mERE (SVO), mERE yields almost the same results in low-resource languages and somewhat higher results in high-resource languages even non-SVO (e.g. Fa, Ar, and Ko) corpora involved during the training stage. Due to the non-SVO languages being very different from others, adding a small amount of such low-resource-language data may be helpful for distinguishing languages in learning language-specific features as submodules in LS are independent. Lastly, we also observe some duplicated F_1 values across low-resource languages. This phenomenon is caused by a small amount of test data.

4.6 Further Analysis

4.6.1 Sentences Concatenation. To validate the effect of the number of sentences for learning the unified features among different languages, we conducted experiments on the different numbers of sentences in concatenation. We learn from Figure 3 that there are evident F_1 improvements with LA on multi-sentence concatenations over only one-single sentence. The multilingual model obtains

the best performance when concatenating with the sentence pair. The increasing number of concatenated sentences has a slight decrease in performance. We conjecture that increasing the number of sentences may also bring somewhat interference.

**Figure 3: The performance of sentences concatenation in the first training stage.**

4.6.2 Selection Mechanism. To observe how the selection mechanism affects our model performance, we also train a one-to-one-based model called mERE14 without using the selection mechanism in the second training stage. Each independent submodule corresponds to a specific language and each sentence is routed via its language prefix. We can visualize from Figure 4 that increasing the

number of parameters also has a great improvement over mERE-LS-LA. However, the mERE14 suffers from the sharp increasing training time and inference time, and big space consumption when the number of languages is large enough. Instead of increasing parameters, our Language-specific Switcher can effectively ameliorate extraction quality with only slight extra parameters and less time consumption since similar languages tend to select the same submodules. The mERE saves nearly 700M model capacity and achieves better performance among most languages compared with mERE14. It is obvious that mERE is light and easy to transfer to other multi-field tasks.

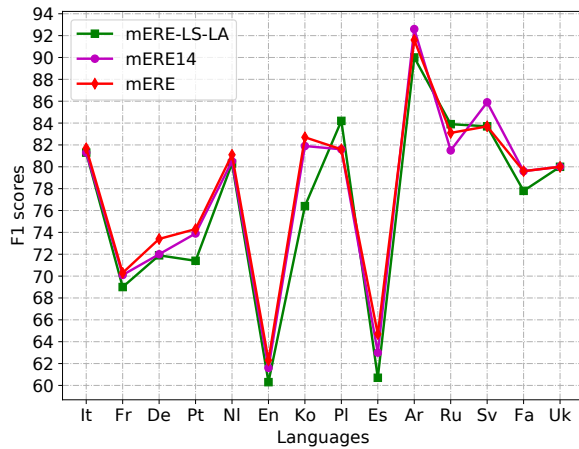


Figure 4: The performance of three models on 14 languages.

5 CONCLUSION

In this paper, we introduce a two-stage training method and a robust framework called mERE for multilingual entity and relation extraction, which ameliorates the sentence representation quality and mitigates the language interference among multiple languages. Specifically, we first learn the generalities across all languages to obtain the unified language representation via the Language-universal Aggregator and then learn the specialties of each language via the Language-specific Switcher. Experimental results demonstrate that our method significantly outperforms both monolingual and multilingual entity and relation extraction baselines, which demonstrates that our framework can effectively extract relational triples among various languages.

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